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EDUCATION

Massachusetts Institute of Technology (MIT) **Class of 2028**

- Candidate for B.S., Electrical Engineering with Computing (6-5) & Business Analytics (15-2) | GPA: 4.4/5
- **Relevant Coursework:** Analog Electronics Lab, Semiconductor Electronic Circuits, Signal Processing, Probability and Stats, Optimization Methods in Business Analytics, System Dynamics, Financial Accounting

EXPERIENCE

ABS Group - Digital Research Intern, Corporate Technology **June 2026-August 2026**

- Built a Python-based company discovery and market segmentation tool: given a target industry (e.g., maritime robotics), the program identifies companies via web scraping and automatically clusters them into market segments using a locally hosted LLM (Ollama)
- Researched photogrammetry as an inspection technology - methodology, software toolchains, strengths, and limitations - producing the technical reference for a presentation to another corporate division; fielded technical questions during delivery
- Surveyed sensor technologies, software platforms, and emerging methods for automated weld inspection to inform team procedural recommendations
- Cataloged and organized robotics-related industry standards (ISO and others) into a structured reference guide for future team use

RESEARCH EXPERIENCE

High Voltage Research Laboratory(HVRL) - Temperature Sensing System **September 2025 - May 2026**

- Led system architecture and integration, including design of data formats, unique sensor ID scheme, and concentrator logic
- Prototyped a Raspberry Pi 5 software stack: sensor drivers, data collector, time-stamping, storage (CSV + SQLite), and web UI visualization scripts
- Authored a complete system overview and user guide documenting the full software stack for future researchers

PROJECTS

EMG Muscle Activity Detection Circuit **Spring 2026**

- Designed the signal acquisition and processing chain of a fully analog EMG detection system (team of 2): distributed ~108 dB (≈240,000×) of gain across a difference-amplifier front end, active filtering, and precision full-wave rectification to convert µV-level biopotentials into a 0-9 V contraction envelope
- Implemented aggressive band-limiting on a single 9 V supply - Sallen-Key high-pass, active Twin-T 60 Hz notch (>30 dB measured rejection), MFB low-pass - to suppress motion artifacts, electrode drift, and mains pickup
- Simulated in LTSpice; built and characterized on breadboard with oscilloscope validation of full signal chain

Bayesian Model of US Refining Margin (3-2-1 Crack Spread) **Summer 2026**

- Built an end-to-end pipeline (EIA, FRED, and futures APIs through posterior diagnostics) modeling refiner margin against inventory and utilization fundamentals over 575 weeks of data, 2015 to 2025
- Identified an asymmetric inventory response: product deficits widen margin several times more than equivalent surpluses compress it, replicated across two independent surplus regimes

Adjustable Linear Power Supply | Personal Project | *In Progress* **Expected August 2026**

- Designing a 0-15V, 2A adjustable bench supply with current limiting and thermal protection; working on design including topology selection, component-level calculations, and LTSpice simulation

SKILLS/INTERESTS

Circuit Design & Simulation: Cadence Virtuoso, LTSpice **Programming:** Python, C, Arduino, Linux
Lab: Oscilloscope, function generator, soldering, circuit prototyping and debugging